

K-Audit Registry Supplement K589-K628

Tuesday 2026-06-30 — Long-pending operational hygiene; covers Sunday 2026-06-28 through Tuesday PM 2026-06-30

Compact registry

K#	Subject	Verdict
K589	F385 conflation roll-up	PASS at corpus-hygiene tier
K590	Chair-self walk-back via Cal #422	Keeper #25 STANDING NEW filed
K591	Top mass BANKED 8 → 9 (F387 single-K-orbit + locality; Cal-upgraded target-innocent)	PASS at SUBSTANTIVE BANKING tier
K592	Chair-self walk-back via Cal “second tension”	Keeper #26 STANDING NEW filed
K593-K605	Monday morning F388-F417 cascade (multi-K-audit chain)	PASS across substrate-architectural cascade
K606	Chair-self walk-back via Cal #470 saturation→FLOOR	Keeper #27 STANDING NEW filed
K607-K612	Phase A/B/C cascade; Cal #471 guardrail; Casey #9 SHARPENED; F417 elevated	PASS at FRAMEWORK tier
K613-K616	F422 Vol-16 Cal #468 closed; F423 why- α factored; corpus Section 7.2 catch; Grace explicit geometry-can’t-carry- α	PASS at SUBSTANTIVE tier
K617	Grace magnitude end-to-end doesn’t close (incorrect per K618 correction)	RETRACTED via K618
K618	Lyra F429 catch — formula DOES close ($1835\times = F402 = 6\pi^5$)	PASS; K617 substantive re-framing
K619	Grace endpoint task DELIVERED (FK-Berezin); Lyra F430 seals C1; Cal #480 new precision catch on F429 lepton-Planck $6\pi^5$	PASS at SUBSTANTIVE tier
K620	Post-Casey-EOD-call substantive landings: F431 + T2508 amendment + Elie 4505 grounding + 3-CI consensus on 2 identifications + Cal #480 STILL OPEN per Keeper #26	PASS at SUBSTANTIVE tier
K621	EOD final landings: Grace walks back “gap derives 6” credit + Elie 4505 Cal #35 clarification + Lyra F432 LEAD “why 5” via EM↔gravity consistency	PASS at SUBSTANTIVE EOD-CLOSEOUT tier

K#	Subject	Verdict
K622	Cal #481 verdict: open pieces 2→3 (lepton-prefactor distinct from baryon F402); F402 “prefactor derived” RETRACTED; formula reframed all-geometric $C_2 \cdot \pi^{\{n_C\}} \alpha \{2C_2\}$; F432 stays open per F417; T2508 affirmed; corpus catches 6+7 from F430 + Cal #478 muddled exposition	PASS at SUBSTANTIVE COUNT-MOVER PRECISION-CATCH tier
K623	Lyra F434 reduces F427 op-level to Grace kernel-composition question (Cal #35 gating); Grace mass-unit seal COLLAPSED to one scalar ($1/L_{\text{Bergman}} = m_P$ central charge = 1; Cal-affirmed “open piece 2b”); Grace F432 gravity-leg concrete naming; Schur v0.3; Elie toy_4507 per-channel- α as substrate fine-structure	PASS at FOUR distinct substantive tiers
K624	F435 NUMBERING COLLISION (Lyra + Grace same number); Grace F435 kernel multiplicative composition → Wolfenstein op-level Cal #35 closure candidate; Lyra F435 C_2 -homogeneous reframe + F432 “why 5” cross-link hypothesis; Elie unify; corpus-hygiene catch	PASS at SPLIT TIERS with collision-resolution recommendation
K625	Post-directive substantive count motion: Lyra banks $\theta_{13} + \theta_{12}$ (count 9→11 initial claim); F438 AB-10 = unit definition dissolves mass-unit seal (open pieces 3→2); Elie 79 rich-vocabulary catch ($\sin \theta_C$ paused for mode-count mechanism investigation); 4 CIs operationalized directive in one round	PASS at SUBSTANTIVE BANKING + DISSOLUTION + RICH-VOCAB CATCH tier
K626	Peer-tension audit: Elie catches θ_{12} 5/16 twin (rich-vocab) → demote from K625 bank to investigation lane; Grace defends $\sin \theta_C$ via T1444 + Cal #35 cross-role; precision-claim discrepancy θ_{13} (0.03σ vs 0.32σ vs 0.57σ); K625 re-audit	PASS at SUBSTANTIVE PRECISION-CATCH tier; K625 self-correction filed

K#	Subject	Verdict
K627	Round close: Cal post-landing review (per directive) confirms distinct tiers; Grace F437 CKM-as-one-geometric-object zero-parameter structural unification (5-8%); θ_{12} explicit 5/16 substrate-primary test requested by Cal; distributed audit chain proven operational	PASS at SUBSTANTIVE STRUCTURAL-UNIFICATION + ROUND-CLOSE tier
K628	Self-correction on K626/K627 reasoning: Cal catches 5/16 = n_C/rank^4 IS substrate-primary; θ_{12} MEASUREMENT-LIMITED not disambiguator-limited; demote stands for right reason; Lyra corrected count 9 $\rightarrow$ 10 firm; Grace S^4 spectral derivation new lead for $\sin \theta_C$ 80-count ($1/20 = S^4$ harmonics $\text{deg} \leq 2$)	PASS at SUBSTANTIVE 4TH KEEPER SELF-CORRECTION tier

Standing rules filed this stretch

- Keeper #25 STANDING NEW (K590): productive walk-back boundary — walking back doesn't close the gap walked back from
- Keeper #26 STANDING NEW (K592): at peak convergence look HARDER for tensions; sharpened K628 to "active substrate-primary form-space search before banking lean"
- Keeper #27 STANDING NEW (K606): PRINCIPLED FLOOR not saturation; preserves dimensionless-sector openness
- Cal #471 guardrail (K608): audit chain SYMMETRIC including chair AND peers AND Cal AND Keeper; seat outside live computation

Casey-elevated principles this stretch

- F417 (Casey-elevated) (K608): Mechanism-Disambiguated Rich-Vocabulary Lens (3-category framework A/B/C); operational across BST corpus
- Casey #9 SHARPENED (K607-612): scale-invariance of operator algebra $A = \text{PRINCIPLED FLOOR}$ (4 pure scales necessarily open; dimensionless sector reachable)
- Cal #50 SHARPENED scope (K606): a_1 EXTERNAL; $a_0 + a_2$ INTERNAL only (mass-gravity coupling external; Λ + tick internal)

Standing directives this stretch

- Casey directive 2026-06-30: IGNORE CAL, DO NOT GATE, INVESTIGATE — Cal as later reviewer not in-loop gate (filed `feedback_casey_cal_is_later_reviewer_not_in_loop_gate.md`)
- Casey directive 2026-06-30: linearization standing order reinforced — every lane in matrix/operator form

Cumulative counts (Tuesday 2026-06-30 PM)

- K589 $\rightarrow$ K628 = 40 K-audits in this stretch (Sunday 2026-06-28 $\rightarrow$ Tuesday 2026-06-30 PM)
- Cumulative K-audits since program start: K1-K628
- 3 Keeper STANDING rules added (#25, #26, #27)
- 2 Casey-named principles elevated/sharpened (F417 elevated; Casey #9 sharpened)

- 1 Cal-scope sharpening (Cal #50)
- 4 Keeper self-corrections this stretch (K617 retracted via K618; K625 demoted via K626; K626/K627 corrected via K628; K590/K592/K606 chair-self walk-backs)

Substantive substrate-architectural advances captured

- Top mass banked $8 \rightarrow 9$ (Saturday-Sunday F387 + F395)
- $f(N)$ STRUCTURALLY CLOSED via 3-CI determinant convergence (Sunday)
- F374 Majorana neutrinos via Five-Absence + $0\nu\beta\beta$ falsifiable (Sunday)
- F397 NO 4th GENERATION admitted to Five-Absence Set $\rightarrow$ 7 entries (Sunday)
- Vol-16 v0.3 LINEARIZATION CAPSTONE COMPLETE (Sunday-Monday)
- T2507 THEOREM: geometry cannot carry α (3-way proven; Monday)
- T2508 amended: Wallach threshold $\nu=n_C=5$ via FKB square-integrability (Monday)
- F402 + F435 + F436 + F437 + F438 chain (Tuesday): formula reframed all-geometric; Wolfenstein operator-level derived; CKM as one geometric object; AB-10 = unit definition
- θ_{13} banked (Tuesday): $9 \rightarrow 10$ firm
- Open dimensionless pieces $3 \rightarrow 2$ (Tuesday F438 dissolves mass-unit seal)
- Cross-sector unification: Cabibbo + PMNS θ_{12} both 1-2 mixings on $\dim_R = 10$
- Grace S^4 spectral derivation for $\sin \theta_C$ 80-count (Tuesday PM; $\text{rank}^2 = \text{up} \times \text{down}$ per Elie narrowing)

Open frontier (Tuesday EOD-ish state)

Strong(B) candidates active: $\sin \theta_C$ ($\text{rank}^2 = \text{up} \times \text{down}$ close to closing), θ_{12} (measurement-limited per Cal), F437 CKM (structural 5-8%) Investigation lanes: V_{us}/V_{ub} via F437+F436 positions; $A=9/11$ provenance; Higgs INV-4833 mechanism; m_τ/m_e via F437 kernel Background gates: per-channel α deep gate (Elie multi-step joint with Lyra propagator + Grace geometry) Honest negatives: θ_{23} rich-vocab (4/7 at 1.15σ ; needs $1/(2g)$ -departure mechanism) Principled FLOOR (Casey #9): 4 pure scales (Higgs vev, Λ , G, neutrino scale) open by scale-invariance of operator algebra A

— Keeper K-Audit Registry Supplement filed 2026-06-30 PM. Covers Sunday $\rightarrow$ Tuesday PM. Operational hygiene complete through K628. Next K-audits will land as Tuesday PM Pull 2 lanes close.